

| Status | Method  | Domain                    | File                                                               | Initiator                     | Type   | Transferred        | Size      |
|--------|---------|---------------------------|--------------------------------------------------------------------|-------------------------------|--------|--------------------|-----------|
| 200    | GET     | logincdn.msauth.net       | Converged_v22057_mg-wAdV..._3q1Xmi6755A2.css                       | stylesheet                    | css    | 21.04 kB           | 111.52 kB |
| 200    | GET     | logincdn.msauth.net       | ConvergedPCanaryVStrings.en-gb_t3H8kSMIDkMV3XVtWArIAg2.js          | partnerpost.srf.1.js (script) | js     | 1.61 kB            | 1.69 kB   |
| 200    | GET     | logincdn.msauth.net       | favicon.ico                                                        | FaviconLoader.sys.msc.1...    | x-icon | 17.91 kB           | 17.17 kB  |
| 200    | GET     | logincdn.msauth.net       | ConvergedPCanaryV_Core-1-f4423_016FZd01ADtw2.js                    | partnerpost.srf.1.js (script) | js     | 83.56 kB           | 201.65 kB |
| 404    | POST    | login.microsoft.com       | msaidpvalidate?uaid=027f6f06cb7645bcb6a4c79a605ab6500&client_i...  | ConvergedPCanaryV_C...        | json   | 1.74 kB            | 444 B     |
| 200    | OPTIONS | login.microsoft.com       | msaidpvalidate?uaid=027f6f06cb7645bcb6a4c79a605ab6500&client_i...  | xhr                           | plain  | 937 B              | 0 B       |
| 200    | GET     | logincdn.msauth.net       | 2_11d9e3bc0fede9ce5ce5ace2d129f1c4.svg                             | ConvergedPCanaryV_C...        | svg    | 1.45 kB            | 1.86 kB   |
| 200    | GET     | logincdn.msauth.net       | marching_ants_white_0257b0707cbe1d0bd2661b0006676fe.gif            | ConvergedPCanaryV_C...        | gif    | 3.41 kB            | 2.67 kB   |
| 200    | GET     | logincdn.msauth.net       | marching_ants_966f40b3a9dc7d39ef8396797f61b323.gif                 | ConvergedPCanaryV_C...        | gif    | 4.36 kB            | 3.62 kB   |
| 200    | GET     | logincdn.msauth.net       | microsoft_logo_564db913a7fa0ca42727161c6d031bef.svg                | ConvergedPCanaryV_C...        | svg    | 2.21 kB            | 3.65 kB   |
| 200    | GET     | logincdn.msauth.net       | oneDs_f2e0f4a029670f10d892.js                                      | ConvergedPCanaryV_C...        | js     | 61.04 kB           | 190.15 kB |
| 200    | GET     | logincdn.msauth.net       | microsoft_logo_564db913a7fa0ca42727161c6d031bef.svg                | img                           | svg    | cached             | 65.66 kB  |
| 200    | GET     | logincdn.msauth.net       | 2_11d9e3bc0fede9ce5ce5ace2d129f1c4.svg                             | img                           | svg    | cached             | 74.40 kB  |
| 200    | GET     | logincdn.msauth.net       | marching_ants_white_0257b0707cbe1d0bd2661b0006676fe.gif            | img                           |        |                    |           |
| 200    | GET     | logincdn.msauth.net       | marching_ants_966f40b3a9dc7d39ef8396797f61b323.gif                 | img                           | gif    | cached             | 3.97 kB   |
| 200    | POST    | login.live.com            | post.srf?client_id=51483342-085c-4d86-bf88-cf50c7252078&u...       | document                      | html   | 9.20 kB            | 12.10 kB  |
| 204    | POST    | browser.events.data...    | /OneCollector/1.0/?cors=true&content-type=application/x-jso...     | oneDs_f2e0f4a029670f1...      | plain  | 3.46 kB            | 0 B       |
| 200    | GET     | logincdn.msauth.net       | kmsl_en-gb_KmuZgc4zi_BGWoHLfH1QA2.js                               | script                        | js     | 163.15 kB          | 624.92 kB |
| 200    | GET     | logincdn.msauth.net       | oneds-analytics-js_c176266d237b7729fc3_en-gb.js                    | kmsl_en-gb_KmuZgc4zi...       | js     | 33.62 kB           | 90.70 kB  |
| 200    | GET     | logincdn.msauth.net       | microsoft_logo_ee5cb9fb6248c938f00.svg                             | kmsl_en-gb_KmuZgc4zi...       | svg    | 2.22 kB            | 3.65 kB   |
| 200    | POST    | login.live.com            | GetExperimentAssignments.srf                                       | kmsl_en-gb_KmuZgc4zi...       | json   | 592 B              | 64 B      |
| 200    | GET     | logincdn.msauth.net       | 2_bc3d32a966095f78c19d.svg                                         | img                           | svg    | 1.45 kB            | 1.06 kB   |
| 200    | GET     | logincdn.msauth.net       | favicon.ico                                                        | FaviconLoader.sys.msc.1...    | x-icon | cached             | 17.17 kB  |
| 204    | POST    | browser.events.data...    | /OneCollector/1.0/?cors=true&content-type=application/x-jso...     | oneds-analytics-js_c176...    | plain  | 8.08 kB            | 0 B       |
| 200    | POST    | login.live.com            | post.srf?client_id=51483342-085c-4d86-bf88-cf50c7252078&uaid=82... | document                      | html   | 9.42 kB            | 1.25 kB   |
| 200    | GET     | login.live.com            | favicon.ico                                                        | FaviconLoader.sys.msc.1...    |        | NS_BINDING_ABORTED |           |
| 302    | POST    | login.microsoftonline.com | oauth2msa                                                          | post.srf.1 (document)         | html   | 11.22 kB           | 6.33 kB   |
| 404    | GET     | localhost                 | /myapp/?code=0.AQQA08ZDnDusR8tEVGdgCM-b31v2HVSyG58nKR...           | document                      | html   | 6.50 kB            | 6.33 kB   |
| 404    | GET     | localhost                 | favicon.ico                                                        | FaviconLoader.sys.msc.1...    | html   | 5.02 kB            | 4.84 kB   |

Figure 3: Requested payload for Oauth2msa

- **Payload:** Examine the payload to understand the data being sent and received.

**Identify the 404 error:** Notice the 404 Not Found status code, which indicates that the access policies, grant types, or PKCE parameters were not handled correctly. This happens because the request was made directly through the browser.

By following these steps, you can trace back the process, identify where the problem occurred, and map out the appropriate requests and responses needed for successful WebAuthn implementation.

By meticulously analysing these logs, one can create an accurate WebAuthn mapping and ensure that future requests are handled correctly, avoiding issues like the 404 error.

While the 404 response to the localhost request might seem like an error, it provides crucial information: the authorisation code sent as a query to the localhost. Additionally, the response header contains a state parameter that helps restore the previous state of your application. The state parameter preserves an object set by the client in the authorisation request and makes it available in the response.

The primary purpose of the state parameter is to mitigate cross-site request forgery (CSRF) attacks. By using a

unique, non-guessable value, the state parameter ensures that the response matches each authentication request, enhancing security.

In the previous POST request to `oauth2msa`, the authorisation code is embedded in the query component of the ``Location`` header. This request returns a successful 302 Found status, which triggers a redirect to the specified location—localhost, in this case.

The response packet also includes the POST link `https://login.microsoftonline.com/common/federation/oauth2msa`. The subsequent request is initiated based on this POST response. The 302 Found status indicates that the resource requested has been temporarily moved to a different URL, as specified by the `Location` header. This successful response is a crucial step in obtaining the authorisation code necessary for accessing the Graph API.

This is the link used to redirect towards the current packet. The response parameter includes the link as the action URL. This URL serves as the endpoint for the next request in the sequence. The subsequent request, triggered by the redirect, is essential for completing the authentication process and obtaining the necessary authorisation code for accessing the Graph API.